

1 Number Game :

```
import java.util.*;
```

```
public class NumberGuessingGame {
```

```
    private static final Scanner scanner = new Scanner(System.in);
```

```
    private static final Random random = new Random();
```

```
    // Leaderboard: store attempts and player number
```

```
    private static final List<Integer> leaderboard = new ArrayList<>();
```

```
    public static void main(String[] args) {
```

```
        System.out.println("🎯 Welcome to the Number Guessing Game!");
```

```
        int totalScore = 0, gamesPlayed = 0, bestScore = Integer.MAX_VALUE;
```

```
        boolean playAgain = true;
```

```
        while (playAgain) {
```

```
            int maxAttempts = chooseDifficulty();
```

```
            if (maxAttempts == 0) { // Exit option chosen
```

```
                break;
```

```
            }
```

```
            int maxRange = chooseRange();
```

```
            long startTime = System.currentTimeMillis();
```

```

int attemptsUsed = playRound(maxAttempts, maxRange);
long timeTaken = System.currentTimeMillis() - startTime;

gamesPlayed++;
if (attemptsUsed > 0) { // player guessed correctly
    int score = calculateScore(maxAttempts, attemptsUsed);
    totalScore += score;
    bestScore = Math.min(bestScore, attemptsUsed);
    updateLeaderboard(attemptsUsed);

    System.out.println("🕒 Time taken: " + (timeTaken / 1000.0) + "
seconds");

    System.out.println("🏆 You earned " + score + " points this round!");
}

playAgain = askPlayAgain();
}

printSummary(gamesPlayed, totalScore, bestScore);
printLeaderboard();

System.out.println("Thanks for playing! 🙌");
}

private static int chooseDifficulty() {
    while (true) {
        System.out.println("\nChoose difficulty:");
    }
}

```

```
System.out.println("1. Easy (10 attempts)");
System.out.println("2. Medium (7 attempts)");
System.out.println("3. Hard (5 attempts)");
System.out.println("4. Exit");
System.out.print("Enter choice: ");
```

```
int choice = getValidInteger();
switch (choice) {
    case 1: return 10;
    case 2: return 7;
    case 3: return 5;
    case 4: return 0; // Exit
    default:
        System.out.println("⚠ Invalid choice! Please select 1-4.");
}
}
}
```

```
private static int chooseRange() {
    System.out.println("\nChoose number range:");
    System.out.println("1. 1 to 50");
    System.out.println("2. 1 to 100");
    System.out.println("3. 1 to 200");
    System.out.print("Enter choice: ");

    int choice = getValidInteger();
```

```
switch (choice) {  
    case 1: return 50;  
    case 3: return 200;  
    default: return 100;  
}  
}
```

```
private static int playRound(int maxAttempts, int maxRange) {  
    int targetNumber = random.nextInt(maxRange) + 1;  
    int attempts = 0;
```

```
    System.out.println("\nI have selected a number between 1 and " +  
maxRange + ". Can you guess it?");  
    while (attempts < maxAttempts) {  
        System.out.print("Enter your guess: ");  
        int userGuess = getValidInteger();  
        if (userGuess < 1 || userGuess > maxRange) {  
            System.out.println("⚠ Please guess a number within the range 1 to " +  
maxRange + ".");  
            continue;  
        }  
        attempts++;  
  
        if (userGuess == targetNumber) {  
            System.out.println("🎉 Congratulations! You guessed the number " +  
targetNumber + " in " + attempts + " attempts.");  
            return attempts; // success
```

```
    } else if (userGuess < targetNumber) {  
        System.out.println("Too low! " + giveHint(userGuess, targetNumber));  
    } else {  
        System.out.println("Too high! " + giveHint(userGuess, targetNumber));  
    }  
}
```

```
    System.out.println(" ❌ Sorry, you've run out of attempts. The correct  
number was " + targetNumber + ".");  
    return -1; // failure  
}
```

```
private static String giveHint(int guess, int target) {  
    int diff = Math.abs(guess - target);  
    if (diff > 20) return "You're way off!";  
    else if (diff > 10) return "You're getting closer!";  
    else return "Very close!";  
}
```

```
private static int getValidInteger() {  
    while (true) {  
        try {  
            return Integer.parseInt(scanner.nextLine().trim());  
        } catch (NumberFormatException e) {  
            System.out.print("⚠ Invalid input! Please enter a number: ");  
        }  
    }  
}
```

```
}  
}
```

```
private static boolean askPlayAgain() {  
    while (true) {  
        System.out.print("Do you want to play again? (yes/no): ");  
        String input = scanner.nextLine().trim().toLowerCase();  
        if (input.equals("yes") || input.equals("y")) return true;  
        else if (input.equals("no") || input.equals("n")) return false;  
        else System.out.println("⚠ Please answer 'yes' or 'no'.");  
    }  
}
```

```
private static int calculateScore(int maxAttempts, int attemptsUsed) {  
    // Score multiplier based on difficulty  
    int baseScore = 10;  
    double multiplier = 1.0;  
    if (maxAttempts == 10) multiplier = 1.0;    // Easy  
    else if (maxAttempts == 7) multiplier = 1.5; // Medium  
    else if (maxAttempts == 5) multiplier = 2.0; // Hard  
  
    // More points for fewer attempts  
    int score = (int) (baseScore * multiplier * ((double)(maxAttempts -  
attemptsUsed + 1) / maxAttempts));  
    return Math.max(score, 1); // at least 1 point  
}
```

```
private static void updateLeaderboard(int attempts) {  
    leaderboard.add(attempts);  
    Collections.sort(leaderboard);  
    if (leaderboard.size() > 3) {  
        leaderboard.remove(leaderboard.size() - 1); // keep top 3 only  
    }  
}
```

```
private static void printSummary(int gamesPlayed, int totalScore, int  
bestScore) {  
    System.out.println("\n 🎮 Game Over!");  
    System.out.println("Total games played: " + gamesPlayed);  
    System.out.println("Your total score: " + totalScore);  
    if (bestScore != Integer.MAX_VALUE) {  
        System.out.println("Your best performance: guessed in " + bestScore + "  
attempt(s).");  
    }  
}
```

```
private static void printLeaderboard() {  
    System.out.println("\n 🏆 Leaderboard (Top 3 best attempts):");  
    if (leaderboard.isEmpty()) {  
        System.out.println("No scores yet.");  
    } else {  
        for (int i = 0; i < leaderboard.size(); i++) {  
            System.out.println((i + 1) + ". " + leaderboard.get(i) + " attempt(s)");  
        }  
    }  
}
```

}
}
}
}